

Marina Vemmou

School of Computer Science, Georgia Institute of Technology, Atlanta, GA, USA

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SUMMARY

Computer architect specializing in hardware-software co-design for heterogeneous memory and compute systems. My research targets performance and efficiency bottlenecks at the boundaries between compute, memory, and networking, with a focus on enabling massive parallelism and accelerating networking stacks and latency-sensitive services. My work spans microarchitecture simulation, performance profiling, and systems software, applied to problems in memory subsystem design, network I/O, and emerging interconnects like CXL. Broadly interested in CXL and disaggregated memory hierarchies, and datacenter energy efficiency.

EDUCATION

Georgia Institute of Technology (Georgia Tech) Fall 2019–Summer 2026

PhD in Computer Science, Track: Computer Architecture, School of Computer Science

Dissertation: Toward Adaptive Modular Computing Systems Through More Expressive And Elastic Hardware and Software Interfaces

Advisor: [Assoc. Prof. Alexandros Daglis](#)

GPA: 4.0/4.0

National Technical University of Athens (NTUA) Fall 2013–Spring 2019

5-year joint BSc & MSc degree in Electrical and Computer Engineering, School of Electrical and Computer Engineering

Diploma Thesis: Application classification techniques' design for interference mitigation in multiprocessor systems

Advisor: [Prof. Georgios Goumas](#)

GPA: 9.20/10

RESEARCH EXPERIENCE

School of Computer Science, Georgia Tech, Graduate Research Assistant Fall 2019–Spring 2026

1. High-Throughput Synchronous Persistence (COSPlay) [MICRO'21]

- **Problem**: Crash-consistent applications utilizing persistent memory serialize persist operations conservatively, placing persistent memory latency on the critical path.
- **Insight**: Independent tasks have separate persist contexts the software knows but cannot communicate to hardware under the current interface.
- **Contribution**: Three-layer co-design comprising a coroutine runtime for task switching at persist barriers, interface primitives that convey task associations to hardware, and targeted CPU modifications that overlap persist operations across independent tasks while preserving intra-task ordering.
- **Result**: Up to $7.3\times$ throughput improvement for high-persist-latency workloads.

2. Eliminating Network-Induced Memory Bandwidth Waste (Sweeper) [MICRO'22]

- **Problem**: DDIO-placed network buffers are indistinguishable from live data in the LLC, causing unnecessary writebacks that waste memory bandwidth.
- **Insight**: Applications know when a buffer is fully consumed but have no mechanism to communicate this to the cache hierarchy.
- **Contribution**: Software API for programmers to declare buffer retirement, paired with a hardware cache-line invalidation instruction that discards consumed buffers without triggering a writeback or modifying the memory consistency model.
- **Result**: Boosts throughput by $1.02\text{--}2.6\times$ by eliminating consumed-buffer writebacks.

3. Elastic Miss-Handling for Heterogeneous Memory Systems (MORIA) [MICRO'26]

- **Problem**: Statically sized MSHRs cannot adapt to runtime MLP demands; MSHR exhaustion stalls the cache and blocks the pipeline.
- **Insight**: Outstanding miss counts and underutilized in-cache storage are observable at runtime and can be repurposed for additional miss-handling state.
- **Contribution**: Dynamic miss-handling architecture that repurposes existing in-cache storage to expand MSHR capacity on demand, enabling runtime adaptation to varying MLP requirements without dedicated hardware overhead.
- **Result**: Average $1.14\times$ speedup over a static baseline, with up to $1.98\times$ for individual applications.

NVIDIA Research, Santa Clara, CA, USA, Research Intern Summer 2024

Supervisor: [Dr. Benjamin Klenk](#)

- Investigated whether pooling discrete NICs across multiple GPUs outperforms dedicated, per-GPU NIC attachment for scale-out GPU networking in AI systems.

- Modeled NIC-pooling vs. dedicated-NIC topologies using NVIDIA's internal GPU-networking simulation infrastructure.

Microsoft Research, Redmond, WA, USA, *Research Intern* Summer 2022
Supervisor: [Dr. Sameh Elnikety](#)

- Investigated software techniques to accelerate networking and enforce QoS in containerized datacenter environments.

Computer Systems Laboratory, NTUA Spring 2018–Spring 2019

- Designed and evaluated techniques for online, lightweight application characterization using machine learning.
- Developed classification methods for interference mitigation in multiprocessor systems.

PUBLICATIONS

- **MORIA: Mining the Cache for Miss Parallelism**
[Marina Vemmou](#), Alexandros Daglis
59th IEEE/ACM International Symposium on Microarchitecture (MICRO), 2026
- **Patching up Network Data Leaks with Sweeper**
[Marina Vemmou](#), Albert Cho, Alexandros Daglis
55th IEEE/ACM International Symposium on Microarchitecture (MICRO), 2022
- **COSPlay: Leveraging Task-Level Parallelism for High-Throughput Synchronous Persistence**
[Marina Vemmou](#), Alexandros Daglis
54th IEEE/ACM International Symposium on Microarchitecture (MICRO), 2021
- **High-Throughput Persistence with Coroutines**
[Marina Vemmou](#), Alexandros Daglis
The Third Young Architect Workshop (YArch) at ASPLOS 2021

AWARDS AND SCHOLARSHIPS

Georgia Tech SCS-SCP Shining Star Award Spring 2024
Georgia Tech College of Computing Outstanding Graduate Teaching Assistant Award Spring 2023
Georgia Tech SCS Incubator Graduate Fellowship Fall 2021
Gerondelis Foundation Graduate Study Scholarship Fall 2020

TEACHING EXPERIENCE

School of Computer Science, Georgia Tech, *Teaching Assistant* Fall 2022, Fall 2020
CS 6290 - High Performance Computer Architecture

School of Electrical and Computer Engineering, NTUA, *Lab Assistant* Spring 2018
Operating Systems

MENTORING

Shreyan Siddharth Jabade, MS student, School of Computer Science, Georgia Tech Spring 2024–Fall 2024
Rachana Aithal Karkada Ramesh, MS student, School of Computer Science, Georgia Tech Fall 2023–Spring 2024
Prashant Ramnani, MS student, School of Computer Science, Georgia Tech Fall 2022–Spring 2024
Manish Manchali, MS student, School of Electrical & Computer Engineering, Georgia Tech Spring 2023
Dhruva Dilip Devasthale, MS student, School of Electrical & Computer Engineering, Georgia Tech Spring 2023

SERVICE AND OUTREACH

Association for Graduate Women and Gender Diversity, College of Computing, Georgia Tech ... Summer 2024
Faculty Recruitment Committee, School of Computer Science, Georgia Tech Spring 2024
Graduate Student Association Board, School of Computer Science, Georgia Tech 2023–2024
Shadow PC, EuroSys 2023 Fall 2023

TECHNICAL SKILLS

Programming Languages: C/C++, Python, UNIX/Bash, LaTeX, Assembly (x86, ARM)
Performance Modelling and Monitoring: ZSim, ChampSim, gem5, Intel RDT, PAPI, Pin

LANGUAGES

English
Greek

Full professional proficiency
Native